Research Roles and Related Work — Working Draft

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Abstract

This working draft synthesizes recent literature on Long Chain-of-Thought (Long-CoT) reasoning in large language models. We structure 24 representative studies across four technical research roles: A1 (Training and Data Strategies), A2 (Inference and Test-Time Control), A3 (Analysis and Mechanistic Understanding), and A4 (Evaluation Frameworks and Benchmarks). In addition, we review nine prior survey taxonomies, examining their organizing principles, scoping frameworks, and comparative perspectives. The collection prioritizes direct Long-CoT studies while marking adjacent and foundational evidence.

1 Related Work and Taxonomy Comparison

The surge in reasoning models has prompted surveys organized around cognitive frameworks, search, compute scaling, evaluation, and efficiency. We compare nine representative surveys [4, 8, 12, 15, 21, 22, 25, 31, 33] with our research roles: **A1 Training/Data**, **A2 Inference/Control**, **A3 Analysis**, and **A4 Evaluation/Benchmark**. We cite pinned preprint versions; the publication year or taxonomy of a later version may differ. Our comparison is selective and does not establish priority over every prior survey.

1.1 Organizing Principles of Prior Surveys

The examined surveys address reasoning literature from complementary technical perspectives, as summarized in Table 1.

Reflective Dynamics and Exploration. Chen et al. [4] structure literature around three primary capabilities: *Deep Reasoning* (format design and training paradigms), *Feasible Reflection* (subdivided into *Feedback* for error identification and *Refinement* for corrective trajectory adjustment), and *Extensive Exploration* (inference scaling via internal deliberation and external tool interaction). Separating feedback from refinement provides an effective conceptual distinction between identifying an error and successfully generating a valid revision.

Dual-Process Cognitive Transitions. Li et al. [15] frame reasoning through a System 1 to System 2 cognitive transition blueprint spanning three interrelated dimensions: *Features* (observable output behaviors and reasoning patterns), *Methods* (core algorithmic approaches comprising Structure Search, Reward Modeling, Self Improvement, Macro Action, and Reinforced Fine-Tuning [RFT]), and *Evolution* (developmental progression from intuitive generation to deliberate inference).

Multi-Dimensional Test-Time Scaling (4W). Zhang et al. [31] formalize test-time compute scaling across four fundamental questions:

1. **What to scale**: Parallel sampling (e.g., Best-of-N, self-consistency), sequential refinement (iterative revision, budget forcing), hybrid search, and internal scaling (model-controlled compute allocation learned through training).

2. **How to scale**: Tuning-based approaches (reinforcement learning, process supervision) versus inference-based prompting and algorithmic search scaffolding.

3. **Where to scale**: Distinct task domains categorized into *Reasoning-intensive Tasks* (mathematics, formal logic), *Agentic Tasks* (tool use, multi-step planning), and *Others* (general question answering, domain tasks).

4. **How well to scale**: Evaluation criteria covering *Performance* (accuracy, pass@k), *Efficiency* (token budget, computational cost), *Controllability* (budget compliance, stopping behavior), and *Scalability* (empirical saturation exponents and scaling boundaries). Crucially, §8.2 explicitly discusses open questions regarding termination criteria and the challenge of differentiating trajectory length from logical sufficiency.

Inference as Search. Wei et al. [25] conceptualize reasoning through the lens of heuristic search, structuring methodologies by *Node Representation and Granularity* (categorized into *Trace-based*, *State–Action*, and *Terminal-State* representations), *Evaluation Function* (value networks, process reward models, self-evaluation), and *MCTS Dynamics* (selection, expansion, simulation, backpropagation), alongside standardized compute reporting protocols.

Evaluating Step-by-Step Traces. Lee & Hockenmaier [12] focus specifically on the evaluation of intermediate reasoning derivations, classifying assessments across four primary dimensions: *Factuality* (factual accuracy of premises), *Validity* (formal logical entailment between steps), *Coherence* (discourse organization and flow), and *Utility* (contribution toward the final solution). They further organize evaluators into rule-based systems, intrinsic model-based verifiers, and external judges, supported by meta-evaluation benchmarks.

Deliberate Compute Allocation. Ji et al. [8] analyze test-time scaling by contrasting *System-1 adaptation* (modifications to models, inputs, representations, and output token distributions) with *System-2 deliberate reasoning* (incorporating external feedback loops, tree/graph search policies, and dynamic compute budgeting).

Efficient and Adaptive Reasoning. Qu et al. [21] organize methods across inference, SFT, RL, and pretraining; inference includes length budgeting, system/model switching, and parallel search. Sui et al. [22] instead group interventions by their target: model, reasoning output, or input prompt. Zhu and Li [33] separate training-free prompting, pipelines, decoding, and merging from training-based fine-tuning and RL. Their stated scope excludes general model compression, architectural optimization, and multimodal efficiency. Together these surveys establish that shortening and adapting reasoning already have substantial coverage; our role-based organization complements their lifecycle, intervention-target, and training-requirement axes.

1.2 Relationship to Research Roles (A1–A4)

Rather than superseding prior taxonomies, the fourfold research role framework used in this survey provides a modular, functional lens focused on the primary technical contribution of

Survey	Scope & Organizing Principle	Useful Comparison
Chen et al. (2025) [4]	**Deep Reasoning**: Format, Learning. **Feasible Reflection**: Feedback, Refinement. **Extensive Exploration**: Scaling, Internal, External.	Decouples error identification (*Feedback*) from corrective trajectory steering (*Refinement*).
Li et al. (2025) [15]	**System 2 Blueprint**: Features, Methods, Evolution. **Core Methods**: Structure Search, Reward Modeling, Self Improvement, Macro Action, RFT.	Connects observable cognitive patterns directly to algorithmic techniques across training and search.
Zhang et al. (2025) [31]	**What**: Parallel, Sequential, Hybrid, Internal. **How**: Tuning-based, Inference-based. **Where**: Reasoning-intensive, Agentic, Others. **How Well**: Performance, Efficiency, Controllability, Scalability.	Comprehensive multi-dimensional taxonomy balancing task domains against compute budgets and scaling saturation.
Wei et al. (2026) [25]	**Search Architecture**: Node Representation (Trace-based, State–Action, Terminal-State), Evaluation Function, MCTS Dynamics, Compute Reporting.	Systematizes tree search algorithms, node representation granularities, and credit assignment mechanisms.
Lee & Hockenmaier (2025) [12]	**Criteria**: Factuality, Validity, Coherence, Utility. **Evaluator Types**: Rule-based, Intrinsic, External; Meta-evaluation benchmarks.	Classifies evaluation metrics and verification architectures for step-by-step reasoning traces.
Ji et al. (2025) [8]	**Cognitive Shift**: System-1 adaptation (model, input, representation, output) versus System-2 deliberate reasoning (feedback, search).	Contrasts fast intuitive generation against deliberate search and compute budgeting trade-offs.
Qu et al. (2025) [21]	**Lifecycle**: Inference, SFT, RL, pretraining; inefficiency patterns and future evaluation.	Connects compute allocation at inference (A2) with learned efficiency (A1).
Sui et al. (2025) [22]	**Intervention target**: Model, reasoning output, input prompts; data/compression and evaluation.	Organizes overthinking remedies by where they intervene.
Zhu & Li (2025) [33]	**Training requirement**: Training-free versus training-based concise/adaptive thinking.	Separates deployment controls (A2) from learned reasoning policies (A1).

Table 1: Systematic comparison of prior survey taxonomies on Long Chain-of-Thought and test-time reasoning.

each work: **A1 Training/Data** isolates training objectives and data curation recipes; **A2 Inference/Control** analyzes test-time execution and search control; **A3 Analysis** encompasses analytical, behavioral, and mechanistic methodologies; and **A4 Evaluation/Benchmark** covers evaluation protocols and stress-test benchmarks. In team discussions, functional dependency concepts (such as dependency formation, state carrying, and verification) have also been discussed as complementary descriptive lenses for analyzing long-horizon reasoning trajectories. These lenses serve as parallel organizational tools to structure literature analysis rather than competing paradigms or empirically validated stages.

Our proposed emphasis is the connection between how reasoning is learned, how it is controlled, what empirical analyses explain, and what evaluation evidence supports. These are synthesis choices, not a claim that previous surveys omit analysis or evaluation. Comparisons should separately record final-answer accuracy, process quality, efficiency, and faithfulness, with explicit model, split, sampling, and compute budgets. A token reduction alone does not quantify end-to-end latency or establish causal faithfulness.

2 Research Roles in Long Chain-of-Thought Reasoning

To systematically organize recent literature on Long Chain-of-Thought (Long-CoT) reasoning, we structure research efforts along four primary technical axes:

1. **A1: Training/Data**—optimization objectives, reinforcement learning (RL) recipes, curriculum data curation, and distillation pipelines;
2. **A2: Inference/Control**—dynamic early-stopping, search tree navigation, error trap recovery, and modular context isolation;
3. **A3: Analysis**—discourse structure modeling, counterfactual causality testing, combinatorial scaling limits, and overthinking dynamics;
4. **A4: Evaluation/Benchmark**—long-horizon state-carrying challenges, graph-driven redundancy metrics, and process supervision protocols.

Crucially, these research roles represent technical research perspectives rather than mutually exclusive model categories. While individual studies often span multiple roles, we categorize each paper's *primary role* according to its central technical contribution and assign *secondary roles* to capture cross-cutting aspects. The subgroup divisions within each role represent inductive organizational groupings proposed by this survey synthesis to highlight distinct methodological directions.

Scope and assignment rule. We use Long-CoT to denote extended deliberation involving exploration or verification, without a universal token threshold; long input context alone does not establish this scope. A1 changes training data or model parameters, A2 changes inference execution, A3 primarily investigates a phenomenon, and A4 primarily introduces an evaluation artifact. We include 24 representative studies (6/5/8/5 by primary role), of which 19 directly study Long-CoT, four provide adjacent reasoning evidence, and one supplies foundational faithfulness methodology. Cross-references do not count as additional papers. This is a selective synthesis rather than an exhaustive systematic review; reported results have not been independently reproduced.

2.1 Training/Data (A1)

Research in A1 addresses how long-horizon deliberative reasoning capabilities are established during post-training, spanning pure reinforcement learning, minimalist data-centric supervision, and multi-stage curriculum pipelines.

2.1.1 RL-Driven Long-CoT Emergence and Scale

A prominent line of work investigates the extent to which complex reasoning behaviors can arise directly through large-scale reinforcement learning with rule-based verification. In **DeepSeek-R1-Zero** [6], the policy is optimized directly from the base model `DeepSeek-V3-Base` via Group Relative Policy Optimization (GRPO), which computes advantage scores over sampled output groups without requiring a separate critic network:

$$A_i = \frac{r_i - \text{mean}(\{r_j\})}{\text{std}(\{r_j\}) + \epsilon}, \tag{1}$$

where rewards r_i reflect answer correctness and structural formatting tags. In this experimental setup, DeepSeek-R1-Zero autonomously developed self-verification and extended exploration trajectories, reaching 77.9% Pass@1 on AIME 2024 and 95.9% on MATH-500. However, the authors noted practical challenges including language mixing and poor readability. To address these issues, the finalized **DeepSeek-R1** introduces a multi-stage recipe: cold-start SFT with several thousand curated traces, reasoning RL, rejection sampling creating an 800k synthetic dataset, and multi-task alignment RL, reaching 79.8% on AIME 2024 (671B MoE) and demonstrating successful SFT distillation to smaller dense models (72.6% on Qwen-32B).

Extending RL scaling to long contexts, **Kimi k1.5** [10] scales training contexts up to 128k tokens across text and vision modalities. The authors employ Online Policy Mirror Descent with relative entropy regularization, utilizing a value-head-free policy and incorporating negative gradients from incorrect rollouts. For reward modeling across extended trajectories, Kimi k1.5 trains a Chain-of-Thought Reward Model (CoT-RM) on 800k samples that outputs intermediate rationales before predicting final answer correctness in JSON format, reporting 98.5% accuracy on authors' manual spot checks compared to 84.4% for classic outcome reward models, while achieving 77.5% Pass@1 on AIME 2024. To mitigate test-time latency, the authors introduce the *Long2short* framework, applying shortest-path rejection sampling and length-penalized preference optimization to distill long reasoning knowledge into shorter traces.

2.1.2 Minimalist Data-Centric Supervision

In contrast to massive RL or large-scale distillation corpora, data-centric studies investigate whether targeted, high-quality demonstration sets can elicit reasoning capabilities on capable base models. The **s1** framework [19] curates the `s1K` dataset—1,000 challenging problems paired with reasoning traces distilled from `gemini-2.0-flash-thinking-exp-1219`. Fine-tuning `Qwen2.5-32B-Instruct` on s1K yields 50.0% Pass@1 on AIME 2024 under standard greedy decoding, and reaches 56.7% on AIME 2024 and 59.6% on GPQA Diamond when combined with *budget forcing* (suppressing `</thought>` delimiters and appending "Wait"). This demonstrates high sample efficiency for small curated corpora, though performance remains conditioned on the base model's pre-existing capabilities.

Similarly, **LIMO** [29] filters an 817-sample training set based on problem difficulty and a four-dimensional trace rubric (elaborated reasoning, self-verification, exploration, and adaptive granularity). SFT on `Qwen2.5-32B-Instruct` achieves 63.3% Pass@1 on AIME 2024 and 95.6% on MATH-500. However, the authors report a significant boundary condition: when transferring this minimalist recipe to `Qwen1.5-32B-Chat`, AIME accuracy drops to 9.2% (a 54.1 percentage point drop). This highlights that data-efficient elicitation depends closely on the specific base model evaluated.

2.1.3 Multi-Stage Curriculum Pipelines

Addressing open reproducibility, **Light-R1** [26] documents a curriculum recipe comprising three training stages plus model merging, initialized from an instruction-tuned checkpoint

(`Qwen2.5-32B-Instruct`) rather than pre-training from scratch. The pipeline proceeds through Stage 1 SFT on 76k filtered mathematical problems (with 20k maximum sequence length), Stage 2 SFT on 3,000 competition-grade instances, semi-on-policy DPO over 32k contexts, and weight-space Model Merging. Light-R1-32B shows progressive gains on AIME 2024 from 16.6% (base) to 69.0% (+SFT-1), 73.0% (+SFT-2), 75.8% (+DPO), and 76.6% (+Merging), while GPQA scores fluctuate across stages (64.3% at SFT-1, 60.6% at SFT-2, and 61.8% after merging). Crucially, the authors conduct a benchmark decontamination audit, documenting test-set leakage in earlier open reasoning corpora.

2.1.4 Efficiency-Aware RL and Learned Compute Allocation

Arora and Zanette [1] train existing reasoning models with correctness rewards and a tunable length penalty. On the 7B distilled model at $\alpha = 0.2$, token savings differ substantially by task: 83% on GSM8K versus 27% on AIME 2024. These are efficiency–accuracy trade-offs, rather than accuracy-preserving guarantees for finite training runs. Compression may also reduce measured faithfulness. This direction complements capability-oriented RL by learning when extended deliberation is worth its cost.

2.2 Inference/Control (A2)

Inference control methods govern how computational budget is allocated during generation, addressing overthinking, early error persistence, and long-horizon context overhead.

2.2.1 Dynamic Early-Stopping and Sufficiency Assessment

Long-CoT models frequently exhibit overthinking, continuing to generate verifications and alternative derivations after reaching a sufficient conclusion. To address this without retraining, **DTSR** [27] monitors reflection trigger tokens (e.g., "Wait", "Alternative") to identify evaluation checkpoints, where a metacognitive sufficiency check prompts the model to assess whether the derivation guarantees a conclusive answer. Evaluated on `Qwen3-14B`, DTSR reduces token count from 5,761 to 3,748 (−34.9%) while maintaining accuracy (84.8% vs. 84.4%). However, scale sensitivity is evident: on `Qwen3-8B`, accuracy decreases by 0.9 pp (81.0% vs. 81.9%) to achieve a 32.0% token reduction, showing that self-sufficiency calibration varies across model sizes.

Similarly, **REFRAIN** [23] formalizes stopping criteria based on reflective redundancy. Its discriminator checks for an existing answer hypothesis, a reflection token, and sentence embedding similarity exceeding threshold τ, dynamically tuned via a Sliding-Window UCB bandit. On `gpt-oss-20B`, REFRAIN improves Pass@1 on tested benchmarks (+2.73 pp GSM8K, +3.40 pp MATH-500, +7.07 pp GPQA-Diamond) while reducing token consumption by 33.3% to 44.7%.

2.2.2 Error Trap Detection and Adaptive Recovery

During long-horizon reasoning, early erroneous commitments often exhibit prefix persistence (diagnosed as thinking traps), causing late-stage self-corrections to fail as generation remains anchored to flawed initial premises. **TAAR** [3] analyzes this phenomenon, reporting that 44.4% of diagnosed thinking traps occur within the first 20% of the reasoning sequence. TAAR trains a diagnostic policy π_ϕ to predict trap locations and escape probabilities, triggering *Cut@Trap* to prune the trajectory prior to the trap segment and restart generation with adaptive temperature. With the same reported path budget $K = 4$ on `DeepSeek-R1-Distill-Qwen-8B`, TAAR increases average accuracy across five competition benchmarks from 64.5% (Avg@4) and 66.0% (PRM@4) to 69.1%, with Cut@Trap achieving a 16.5% escape rate. However, on `GPT-OSS-120B`, accuracy slightly decreases (79.2% vs. 80.6%), as stronger frontier models encounter fewer traps and aggressive pruning risks discarding near-correct prefixes.

At a coarser granularity, **MCTSr** [30] integrates Monte Carlo Tree Search with natural language self-refinement, operating over complete mathematical derivations as search nodes. Using a modified UCT selection rule, MCTSr generates textual critiques, produces rewritten solutions, and updates value estimates via self-evaluation. On `LLaMA-3-8B`, an 8-rollout search improves MATH accuracy from 24.36% (zero-shot) to 58.24%, with Level 5 problems advancing from 7.10% to 34.06%.

2.2.3 Modular Context Isolation and State Management

Autoregressive generation over a single flat context incurs quadratic attention cost $O(T^2)$ and context dilution over long sequences. **Recursive Models** [28] address this by formalizing an execution framework where subtasks are dispatched to isolated stack frames via explicit `<call>` and `<return>` control tags. Upon completion, only the concise return value is passed back to the caller while intermediate tokens are cleared from the active context. On SAT problems, a fine-tuned `Qwen2.5-3B-Instruct` recursive model achieves 98% (Easy), 95% (Medium), and 64% (Hard OOD) accuracy, outperforming flat CoT baselines. On 4x4 Go trace generation under length-OOD evaluations, the recursive model retains 38.5% trace accuracy where standard flat CoT drops to 1.0%.

2.2.4 Budget Forcing and Exploration-Depth Control

Two cross-cutting controls complement early stopping: s1's budget forcing [19] regulates deliberation length, while TIP [24] discourages premature thought switching. The latter adds a penalty to transition-token logits within a local decoding window. These controls address different allocation failures: continuing after a sufficient solution versus abandoning a promising branch before exploring it deeply. Their primary paper records remain in A1 and A3, respectively.

2.3 Analysis (A3)

Analytical literature examines internal reasoning behaviors, focusing on discourse structures versus mechanistic causality, graph representations of failure modes, and combinatorial scaling limits.

2.3.1 Discourse Graph Structures vs. Mechanistic Causality

A foundational question in Long-CoT analysis is whether surface discourse links reflect true internal computational dependencies. In **ReasoningFlow** [13], reasoning traces are modeled as Directed Acyclic Graphs (DAGs) with typed discourse edges (*support, attack, verify, assumption*). To test causal dependencies, the authors evaluate *Thought Anchors* (measuring log KL-divergence of step s_j predictions under attention masking of step s_i). On QwQ and AIME, Thought Anchor scores predict ReasoningFlow edges with precision-recall comparable to a simple heuristic baseline selecting neighboring nodes by linear token distance, indicating that alignment between intervention-based scores and discourse edges was limited in that tested setup. In behavioral analysis of observed traces (Figure 5), local verification with explicit error-correction is associated with step-level accuracy of 84% (compared to 64% for prior flawed steps), and corrected steps are four times more likely to yield a correct final answer (20% vs. 5%); these observations reflect behavioral associations across generated traces rather than controlled causal interventions.

Connecting to earlier counterfactual perturbation work, **Lanham et al.** [11] investigate CoT faithfulness across models up to 175B parameters by testing early answering, mistake insertion, and filler tokens. In §2.5.1 and Figure 5, replacing CoT reasoning with uninformative filler tokens (“ ...”) shows no increase in task accuracy across evaluated benchmarks, with slight performance drops on TruthfulQA and OpenBookQA which the authors suggest may stem

from out-of-distribution effects. In their experimental setup, adding test-time tokens without substantive symbolic semantic content did not provide reasoning benefits; this finding applies specifically to the evaluated models and tasks without extrapolating as a universal law for all latent-compute architectures.

2.3.2 Structural Graph Representations and Failure Mode Diagnosis

Moving beyond linear sequence metrics, **CoT2Graph** [2] represents CoT traces as typed dependency graphs, utilizing Graph Neural Networks (GNNs) with GNNExplainer attribution to identify structural signatures. The authors observe that correct traces rely prominently on early strategic planning, whereas failed derivations exhibit overloaded synthesis nodes and unfulfilled dependencies; graph attributions explain the GNN predictor rather than the LLM's internal causal process. On GPQA graduate-level science traces using `DS-Qw14B`, binary prediction of final-answer correctness achieves 65.4% accuracy, outperforming the length-based Logistic Regression baseline (57.6%) by +7.8 percentage points and the tree-structured baseline LCoT2Tree (62.9%) by +2.5 percentage points (Table 1). In Best-of-N ($N = 5$) selection on 600 MMLU-Pro questions with `DS-Qw7B`, CoT2Graph selection achieves 54.5% accuracy compared to the ORM-Best baseline of 50.8%, the strongest non-graph baseline (ORM + uncertainty) of 51.6%, and a 43.2% mean baseline over responses, reaching 56.1% when combined with entropy signals.

Feng et al. [7] compare ten reasoning models and identify the *Failed-Step Fraction* (FSF) as a more consistent predictor of correctness than length or review ratio. FSF-guided selection and controlled removal of abandoned branches support structure-aware control. However, the extracted graph depends on a judge, and an abandoned branch is not necessarily a logically invalid step.

2.3.3 Combinatorial Scaling Limits

Benchmarking deductive reasoning under exponential search spaces, **ZebraLogic** [16] evaluates grid Constraint Satisfaction Problems (CSPs) with search spaces spanning $< 10^3$ to $> 10^9$ candidates. The authors document a *Curse of Complexity*: accuracy drops sharply once the search space exceeds 10^6. Parameter scaling fails to overcome this barrier (`Llama-3.1-405B` achieves 32.6% overall but drops to 1.5% on Large and 0.0% on X-Large). Reasoning models maintain higher accuracy at moderate scale but also degrade severely on X-Large (`o1-full`: 42.5%; `DeepSeek-R1`: 28.5%; `o1-mini`: 12.0%), while repeated sampling (Best-of-N) and self-refinement show limited gains under high combinatorial complexity.

2.3.4 Overthinking, Underthinking, and Self-Doubt

Chen et al. [5] introduce outcome/process efficiency metrics and self-training against overthinking. On MATH500, their QwQ-32B-Preview variant reduces tokens from 2407.9 to 1330.7 while accuracy changes from 93.0% to 92.8%; savings can therefore entail performance loss. Conversely, Wang et al. [24] diagnose frequent switching between shallow attempts. On AIME 2024, TIP increases QwQ-32B-Preview Pass@1 from 38.3% to 44.1%, estimated from 32 samples per problem. A long overall trace can thus coexist with insufficient exploration depth.

Peng et al. [20] analyze the behavioral etiology of overthinking, identifying *self-doubt*—the tendency to repeatedly re-verify already correct derivations—as a primary driver of redundant token consumption. On missing-premise problems where unguided models generate 4,700–6,400 tokens attempting impossible proofs, premise-validation prompts reduce token lengths by 35%–45% while preserving or improving accuracy on tested models.

2.4 Evaluation/Benchmark (A4)

Validating Long-CoT models requires benchmarks that probe long-horizon state tracking, automated metrics that quantify reasoning efficiency, and process supervision protocols.

2.4.1 Long-Horizon Dependency and Scaling Breakdown Benchmarks

Addressing saturated short-form benchmarks, **LongCoT** [18] introduces a benchmark of 2,500 problems generated from expert-designed parameterized templates across five domains (mathematics, chemistry, computer science, chess, and formal logic), with solution lengths varying substantially across problems (typically 10k–100k+ tokens). On 2,000 medium/hard problems, evaluated frontier models achieve modest accuracy: `GPT-5.2` reaches 9.83%, `Gemini 3 Pro` 6.08%, and `Grok 4.1` 2.04%. In a distinct controlled experiment scaling sequential DAG dependencies from 1 to 35 subproblem nodes, accuracy decays faster than the theoretical independent error baseline (p^N), particularly past 15 nodes, supporting the authors' hypothesis of compounding state-carrying decay across extended contexts.

Similarly, **SPaRC** [9] evaluates multi-step spatial planning over text-based 2D grid pathfinding puzzles across five difficulty tiers. Human solvers achieve 98.0% overall accuracy (and 94.5% on Level 5), whereas the strongest evaluated reasoning model (`o4-mini`) achieves 15.8% overall, falling to 1.1% on Level 5 (with over 50% of paths violating movement constraints). Crucially, SPaRC identifies an absence of adaptive compute scaling: while human deliberation scales sharply with problem difficulty, tested models such as `QwQ-32B` emit comparable token volumes across tiers (14,433 tokens on Level 1 vs. 13,114 tokens on Level 5).

2.4.2 Graph-Driven Automated Redundancy Frameworks

To provide automated efficiency metrics without step-level human annotations, **CoTJudger** [14] formalizes an evaluation framework that converts traces into directed dependency graphs and extracts the *Shortest Effective Path* (SEP) using forward depth-first search verified by an LLM consistency judge. This defines the Redundancy Ratio $R = (|V| - L_{\text{eff}})/|V|$, measuring the proportion of inferred graph nodes (reasoning steps) residing outside the SEP. Evaluating 21 reasoning models, CoTJudger finds a substantial proportion of reasoning nodes residing outside the SEP, with redundancy heavily concentrated in repetitive self-verification loops after reaching conclusive states. The authors treat SEP as a judge-dependent structural proxy rather than proof that every excluded step is semantically unnecessary, and observe that distilled models exhibit lower logical density per step compared to teacher models.

2.4.3 Step-Level Process Supervision and Critique Benchmarks

Evaluating process supervision protocols requires granular error localization. **ProcessBench** [32] provides 3,400 test cases with stepwise solutions across four benchmark subsets (GSM8K, MATH, OlympiadBench, Omni-MATH) with expert human annotations identifying the earliest erroneous step. Comparing dedicated Process Reward Models (PRMs) against general LLM critics, ProcessBench shows that PRMs trained on simpler data generalize poorly across diverse test cases: `Skywork-PRM-7B` achieves an average F1 of 42.1% across the four subsets. Conversely, reasoning models prompted as critics perform more strongly: `QwQ-32B-Preview` reaches an average F1 of 71.5%, exceeding `GPT-4o` (61.9%) and trailing dedicated reasoning models such as `o1-mini` (87.9%).

Finally, **CriticBench** [17] establishes a benchmark evaluating the triad of Generation, Critique, and Correction (GQC) across five reasoning domains. Across 17 evaluated LLMs, CriticBench demonstrates a positive correlation between critique accuracy and downstream correction success, while noting that smaller models exhibit conservative self-critique behavior unless guided by external feedback signals.

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